



PROGRAMMING FUNDAMENTALS.
C++ I.

C++ I – Lab 02 Instructions

You are required to submit **two files** for this lab assignment:

◆ 1. Lab02.pdf – Lab Report

This PDF file must include:

1. A **brief explanation** (1–2 sentences) of what the program does.
2. A **screenshot of your full C++ program** as shown in your code editor.
● **Important:** This must be an **image (screenshot)** — *not a copy-paste of the program text*. It should be clear and legible.
3. At least **five screenshots** of your program running with **different input values** for the meal charge.
4. A **comment or caption** for each screenshot stating:
 - The **input value**
 - What is being shown in the output
 - Whether the output is correct

🖼️ Screenshot Guidelines:

- Only show the **relevant part** of the program's input/output console.
- Each screenshot must show **one complete program run**.
- Ensure text is easily readable (minimum **14-point font size** or better).
- Avoid clutter—only include what's relevant to that specific test.

Refer to the **Lab Report Example** provided in the **Blackboard Course Content** area for format and layout guidance.

◆ 2. Lab02.cpp – C++ Program Source Code

This file must contain your complete and properly formatted **C++ program** that solves the following problem:

Programming Challenge: Restaurant Bill

Write a C++ program that:

- Prompts the user to **enter the meal charge** (e.g., 88.67).
 - Calculates the **tax** (6.75% of the meal cost).
 - Adds the tax to get the **subtotal**.
 - Calculates the **tip** (20% of the subtotal).
 - Computes the **total bill** by adding the tip.
 - Displays the following:
 - Meal cost
 - Tax amount
 - Tip amount
 - **Total bill**
-

☒ **Programming Best Practices (from Chapter 2):**

- Use **descriptive variable names**.
- Use **named constants** for tax and tip percentages.
- Include **comments** throughout the code to explain logic.
- Apply proper **indentation and formatting**.

Your program must be tested at least **five times** using **different meal charges**, and the corresponding outputs must be captured in the lab report.

! Important Grading Notes:

- **Failure to comply with all program requirements** or the **lab report format** will result in **lost points**.
 - Incomplete screenshots, unreadable text, missing code images, or failure to follow coding standards will be penalized.
 - Follow instructions **carefully** to ensure full credit.
-

Submission Instructions:

Upload both of the following to **Blackboard** under the **Lab 02** section:

- `Lab02.cpp` – Your C++ source code
- `Lab02.pdf` – Your lab report with all required screenshots and explanations

If you have questions or need help, don't hesitate to email me.

Instructor: LDV
